Ogbonna Arinze

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SUMMARY

Senior Frontend Engineer with 8 years of experience building high-impact web applications across healthcare, fintech, and civic tech. Currently at Genentech (Roche) building clinical trial data platforms with React, Next.js, TypeScript, and GraphQL. Deep expertise in frontend architecture, design systems, data visualization, and performance optimization for complex datasets. Proven track record of implementing robust testing infrastructure with Playwright, establishing CI/CD pipelines with GitHub Actions, and integrating observability tooling including Sentry and Lighthouse to ensure production reliability.

CORE COMPETENCIES

- **Frontend Architecture:** React, Next.js, TypeScript, GraphQL, Vue.js, scalable component systems, design systems, SSR/SSG, and monorepos (Nx, Turborepo).
- **Testing & Quality:** Playwright (E2E in CI/CD), Jest, Cypress, Allure Reports, Storybook, ESLint, Prettier, and CI-driven QA.
- **Performance & Observability:** Core Web Vitals, Lighthouse, Sentry, code splitting, CDN strategies, bundle analysis (webpack-bundle-analyzer, rollup-plugin-visualizer), and React Profiler.
- **Data Visualization:** Recharts, react-chartjs, dynamic dashboards, and large dataset rendering optimization.
- **DevOps & CI/CD:** GitHub Actions, Docker, AWS (CloudFront, S3), Vercel, automated test pipelines, and Nx monorepo orchestration.
- **Backend & Data:** NestJS, NodeJS REST APIs, PostgreSQL, MongoDB, Prisma, GraphQL, JWT authentication, and RBAC.
- **AI & LLM Systems:** RAG pipelines, OpenAI API, LangChain, Vercel AI SDK, vector search (pgvector, Chroma), and embeddings.
- **Leadership:** Technical architecture reviews, mentorship, Agile/Scrum, TDD/BDD, and cross-functional collaboration.

WORK EXPERIENCE

Genentech (Roche) San Francisco, CA (Remote)

Frontend Engineer *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React Chrome extension**, making frontend architectural decisions that optimized workflows for clinical site integrations with upstream EHR systems.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams serving researchers, clinicians, and internal stakeholders.
- Co-developed a full-stack **Next.js** chat widget powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling users to connect data sources and query biomedical data in real time via a conversational interface.
- Optimized application performance by implementing **lazy loading, code splitting**, dynamic imports, and **CDN asset delivery**, reducing load times significantly as measured via **Lighthouse** and **Chrome DevTools**.
- Implemented **JWT-based authentication** and **role-based access control** in **Next.js** with **TypeScript** and **React Query**, using HTTP-only cookies for secure token handling and permission-based UI rendering.
- Introduced **Redux** for state management and **React Query** for **GraphQL** request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline** via **GitHub Actions**, automating reporting with **Allure Reports** on GitHub Pages and configuring **Slack notifications** for build quality visibility.
- Maintained and extended the existing **Vue.js** codebase during the migration period, shipping features and fixing bugs in parallel with the React rebuild to ensure uninterrupted delivery.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality through structured code reviews.

Descasio Lagos, Nigeria (Hybrid)

Senior Frontend Developer *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through **rollup-plugin-visualizer** and **webpack-bundle-analyzer**, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application, significantly boosting speed and user experience.
- Enhanced development workflow by introducing **Jest** and **Enzyme** automated testing frameworks, leading to a **20% decrease** in bug reports and improving release confidence.

- Utilized **Storybook** for documenting React components, streamlining change management and ensuring clear communication of design system patterns across the team.
- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating **CI/CD** processes across the frontend codebase.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across projects.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3, processing thousands to millions of daily transactions and boosting settlement rates by **80%**, demonstrating expertise in **large dataset performance optimization**.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React**, **Redux**, **RxJS** for real-time data streaming, and **Axios** for asynchronous API requests.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing and data management strategies.
- Adopted the native **EventSource API** for real-time data streaming, replacing socket.io for faster updates and a more responsive interface across high-traffic dashboards.
- Built dynamic **data visualization dashboards** using **React** and **react-chartjs**, streamlining management of user roles and permissions with **Redux Toolkit** and **Ant Design**.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to **Heroku** and **AWS**.
- Mentored two interns through frontend and backend projects, providing technical guidance and task management.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 applications**: NestJS API backend, **Next.js** citizen-facing web app with **22 interactive chart types** rendered via **Recharts**, Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service.
- Built the **Next.js** citizen-facing frontend with **React 19**, **Tailwind CSS v4**, and **Recharts**, featuring auto-generated **data visualizations**, dark mode, URL state management, and responsive mobile layouts optimized for large datasets.
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright E2E tests** across 4 test projects (web auth, dashboard auth, desktop Chrome, mobile Pixel 5), and **Docker Compose** production deployment with multi-stage Dockerfiles and health checks.
- Designed the **PostgreSQL 16** database schema with **Prisma v7**, including pgvector extension for vector storage, and integrated **Langfuse** for LLM observability and **Infisical** for secrets management across environments.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*